

Laplace Transform, $\mathcal{L}$

The Laplace Transform of $f(t)$ is

$$\mathcal{L}\{f\} = \int_0^{\infty} f(t) e^{-st} dt = F(s)$$

Example 1

Find Laplace transform of $f(t) = 1$

$$\mathcal{L}\{1\} = \int_0^{\infty} 1 e^{-st} dt$$

$$\text{Let } I_R = \int_0^R 1 e^{-st} dt$$

$$= \int_0^R e^{-st} dt$$

$$= \left[-\frac{1}{s} e^{-st} \right]_0^R$$

$$= \left[-\frac{1}{s} e^{-sR} + \frac{1}{s} \right]$$

$$\mathcal{L}\{1\} = \lim_{R \rightarrow \infty} I_R$$

$$= \lim_{R \rightarrow \infty} \left[-\frac{1}{s} e^{-sR} + \frac{1}{s} \right]$$

$$= \begin{cases} \frac{1}{s}, & s > 0 \end{cases}$$

$$= \begin{cases} \text{Undefined}, & s < 0 \end{cases}$$

Example 2

Find Laplace transform of $f(t) = \cos(at)$

$$\mathcal{L}\{\cos(at)\} = \int_0^{\infty} \cos(at) e^{-st} dt$$

$$\text{Let } I_R = \int_0^R \cos(at) e^{-st} dt$$

$$I_R = \left[-\frac{1}{s} e^{-st} \cos(at) \right]_0^R + \left[\frac{a}{s^2} e^{-st} \sin(at) \right]_0^R - \frac{a^2}{s^2} \int_0^R \cos(at) e^{-st} dt$$

$$I_R = \left[-\frac{1}{s} e^{-sR} \cos(aR) + \frac{1}{s} \right] + \left[\frac{a}{s^2} e^{-sR} \sin(aR) \right] - \frac{a^2}{s^2} I_R$$

$$\left(1 + \frac{a^2}{s^2}\right) I_R = -\frac{1}{s} e^{-sR} \cos(aR) + \frac{a}{s^2} e^{-sR} \sin(aR) + \frac{1}{s}$$

$$I_R = \frac{s^2}{s^2 + a^2} \left[-\frac{1}{s} e^{-sR} \cos(aR) + \frac{a}{s^2} e^{-sR} \sin(aR) + \frac{1}{s} \right]$$

$$\mathcal{L}\{\cos(at)\} = \lim_{R \rightarrow \infty} I_R$$

$$= \frac{s^2}{s^2 + a^2} \left[-\frac{1}{s} e^{-sR} \cos(aR) + \frac{a}{s^2} e^{-sR} \sin(aR) + \frac{1}{s} \right]$$

$$= \begin{cases} \frac{s}{s^2 + a^2}, & s > 0 \\ \text{Undefined}, & s < 0 \end{cases}$$

D	I
+ $\cos(at)$	e^{-st}
- $(-a \sin(at))$	$-\frac{1}{s} e^{-st}$
+ $(-a^2 \cos(at))$	$\frac{1}{s^2} e^{-st}$

Existence of Laplace Transform

A function $f(t)$ has a Laplace Transform if

$$\int_0^{\infty} f(t) e^{-st} dt$$

converges [having a finite value].

1) The function $f(t)$ might grow larger & larger

2) e^{-st} acts as a decay factor when $s > 0$

$\therefore$ For the Laplace Transform to exist, the shrinking part (e^{-st}) must win over $f(t)$ growth.

Sufficient Conditions

1) Piecewise continuous [No vertical asymptotes]

2) Subexponential growth $|f(t)| \leq M e^{\sigma t}$, for $t > T$

~ To check if $f(t)$ can be trapped under a standard exponential curve, $M e^{\sigma t}$

Example 1

Does $\mathcal{L}\{\sinh(t)\}$ exist?

1) No jumps ✓

$$2) \sinh(t) = \frac{1}{2} e^t - e^{-t} \leq \frac{1}{2} e^t \text{ for all } t \geq 0$$

$$\Rightarrow M e^{\sigma t} = \frac{1}{2} e^{(1)t} \Rightarrow |f(t)| \leq M e^{\sigma t} \quad \checkmark$$

Example 2

Does $\mathcal{L}\{e^{t^2}\}$ exist?

1) No jumps ✓

$$2) I = \int_0^{\infty} e^{t^2} e^{-st} dt$$

$$= \int_0^{\infty} e^{t(t-s)} dt \quad \text{variable } t, \text{ constant } s$$

$$\text{Let } t \geq s+1 \Rightarrow t(t-s) \geq t(1)$$

$$\Rightarrow t(t-s) \geq t$$

$$\Rightarrow e^{t(t-s)} \geq e^t$$

$$\int_{s+1}^{\infty} e^{t(t-s)} dt \geq \int_{s+1}^{\infty} e^t dt$$

Since it's known that

$\int_{s+1}^{\infty} e^t dt$ definitely diverges,

and $\int_{s+1}^{\infty} e^{t(t-s)} dt \geq \int_{s+1}^{\infty} e^t dt$,

$\Rightarrow \int_{s+1}^{\infty} e^{t(t-s)} dt$ also diverges.

$\therefore$ Laplace transform doesn't exist for e^{t^2} .

Laplace Transforms Linearity

$$\mathcal{L}\{a f(t) + b g(t)\} = a \mathcal{L}\{f\} + b \mathcal{L}\{g\}$$

Table of Laplace Transforms

$f(t)$	$\mathcal{L}\{f\} = F(s) = \int_0^\infty (f(t) e^{-st}) dt$
1	$\frac{1}{s}$ for $s > 0$
e^{at}	$\frac{1}{s-a}$ for $s > a$
$\sinh(\lambda t)$	$\frac{\lambda}{s^2 - \lambda^2}$ for $s > \lambda $
$\cosh(\lambda t)$	$\frac{s}{s^2 - \lambda^2}$ for $s > \lambda $
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$ for $s > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$ for $s > 0$
t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
t^α for $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$ for $s > 0$
$g(t) e^{at}$	$G(s-a)$
$tg(t)$	$-\frac{d}{ds}(G(s))$
$t^n g(t)$	$(-1)^n \frac{d^n}{ds^n}(G(s))$
$\frac{dy}{dt}$	$sY(s) - y(0)$
$\frac{d^2y}{dt^2}$	$s^2Y(s) - sy(0) - \frac{dy}{dt}(0)$
$\int_0^t (g(\tau)) d\tau$	$\frac{1}{s}G(s)$
$g(t-a)u(t-a)$	$e^{-as}G(s)$
$\delta(t-a)$	e^{-as} for $a \geq 0$
$(g * h)(t) = \int_0^t (g(\tau)h(t-\tau)) d\tau$	$G(s)H(s)$

Example [Linearity]

Find $\mathcal{L}\{\cosh(3t)\}$

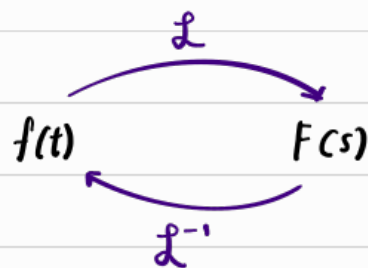
$$\begin{aligned}
 \mathcal{L}\{\cosh(3t)\} &= \mathcal{L}\left\{\frac{1}{2}e^{3t} + \frac{1}{2}e^{-3t}\right\} \\
 &= \frac{1}{2}\mathcal{L}\{e^{3t}\} + \frac{1}{2}\mathcal{L}\{e^{-3t}\} \\
 &= \frac{1}{2}\left[\frac{1}{s-3}\right] + \frac{1}{2}\left[\frac{1}{s+3}\right] \\
 &= \frac{s}{s^2 - 9}
 \end{aligned}$$

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e^{at}	$\frac{1}{s-a}$ for $s > a$
$\sinh(\lambda t)$	$\frac{\lambda}{s^2 - \lambda^2}$ for $s > \lambda $
$\cosh(\lambda t)$	$\frac{s}{s^2 - \lambda^2}$ for $s > \lambda $
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$ for $s > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$ for $s > 0$
t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
t^α for $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$ for $s > 0$
$g(t) e^{at}$	$G(s - a)$
$tg(t)$	$-\frac{d}{ds}(G(s))$
$t^n g(t)$	$(-1)^n \frac{d^n}{ds^n}(G(s))$
$\frac{dy}{dt}$	$sY(s) - y(0)$
$\frac{d^2y}{dt^2}$	$s^2Y(s) - sy(0) - \frac{dy}{dt}(0)$
$\int_0^t (g(\tau)) d\tau$	$\frac{1}{s} G(s)$
$g(t-a) u(t-a)$	$e^{-as} G(s)$
$\delta(t-a)$	e^{-as} for $a \geq 0$
$(g * h)(t) = \int_0^t (g(\tau) h(t-\tau)) d\tau$	$G(s) H(s)$

Inverse of Laplace Transform, $\mathcal{L}^{-1}$

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$



Note:

Linearity holds for inverse

$$\Rightarrow \mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F\} + b\mathcal{L}^{-1}\{G\}$$

Example 1

Find inverse Laplace transform of $F(s) = \frac{2}{s} + \frac{3}{s-1}$

$$\mathcal{L}^{-1}\left\{\frac{2}{s} + \frac{3}{s-1}\right\} = 2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$= 2[1] + 3[e^t]$$

$$= 2 + e^t$$

Example 2 [Partial Fractions]

Find inverse Laplace transform of $F(s) = \frac{2}{(s-2)(s+4)}$

$$\frac{2}{(s-2)(s+4)} = \frac{A}{s-2} + \frac{B}{s+4}$$

$$2 = A(s+4) + B(s-2)$$

$$(0)s + 2 = (A+B)s + (4A-2B)$$

$$\therefore A+B = 0, \quad 2 = 4A-2B$$

$$\Rightarrow A = \frac{1}{3}, \quad B = -\frac{1}{3}$$

$$\therefore F(s) = \frac{1}{3}\left(\frac{1}{s-2}\right) - \frac{1}{3}\left(\frac{1}{s+4}\right)$$

$$\mathcal{L}^{-1}\{F\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\}$$

$$= \frac{1}{3}e^{2t} - \frac{1}{3}e^{-4t}$$

Rule 1: Distinct linear factors in the denominator

$$\text{Example: } \frac{2x+3}{(x+3)(x-6)} \equiv \frac{A}{x+3} + \frac{B}{x-6}$$

Rule 2: Repeated linear factor in the denominator

$$\text{Example: } \frac{x^2}{(x-3)^3} \equiv \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3}$$

Rule 3: Irreducible quadratic factor in the denominator

$$\text{Example: } \frac{3x^2+2x}{(x+2)(x^2+3)} \equiv \frac{A}{x+2} + \frac{Bx+C}{x^2+3}$$

Rule 4: Improper rational functions (Degree of numerator $\geq$ Degree of denominator)

$$\text{Example: } \frac{x^2+3x}{x^2-4}$$

First, perform polynomial division. Then we have

$$\frac{x^2+3x}{x^2-4} = \text{Quotient} + \frac{\text{Remainder}}{x^2-4}$$

$\frac{\text{Remainder}}{x^2-4}$ is a proper rational function which can be expressed in partial fractions using Rule 1, Rule 2 or Rule 3. [Question: In this case, which of the 3 Rules should we use?]

s - shift

$$\mathcal{L}\{g(t)e^{at}\} = G(s-a)$$

Example [s-shift]

Find Laplace Transform of $f(t) = (t^2 + 3)e^{-2t}$

$$g(t) = (t^2 + 3)$$

$$G(s) = \mathcal{L}\{t^2\} + 3\mathcal{L}\{1\}$$

$$= \frac{2}{s^3} + \frac{3}{s}$$

t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
1	$\frac{1}{s}$ for $s > 0$

$$F(s) = G(s-a)$$

$$= G(s - (-2))$$

$$= G(s+2)$$

$$= \frac{2}{(s+2)^3} + \frac{3}{(s+2)}$$

$g(t)e^{at}$	$G(s-a)$
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Inverting using s-shift

Example

Find the inverse transform of $F(s) = \frac{12}{(s-4)^4}$

$$\text{Let } \hat{s} = s - 4 = s - a \Rightarrow a = 4$$

$$G(\hat{s}) = \frac{12}{(\hat{s})^4}$$

$$G(s-4) = \frac{12}{3!} \cdot \frac{3!}{(\hat{s})^{3+1}}$$

$$g(t) = \frac{12}{6} \cdot [t^3]$$

$$= 2t^3$$

t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
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$$f(t) = g(t)e^{at}$$

$$= 2t^3 e^{4t}$$

$g(t)e^{at}$	$G(s-a)$
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Derivative Property of Laplace

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} [F(s)]$$

Example

Find $\mathcal{L}\{t \sin(\omega t)\}$

$$\mathcal{L}\{t \sin(\omega t)\} = (-1)^1 \frac{d}{ds} [\mathcal{L}\{\sin(\omega t)\}]$$

$$= - \frac{d}{ds} \left[\frac{\omega}{s^2 + \omega^2} \right]$$

$$= - \frac{0(s^2 + \omega^2) - \omega(2s)}{(s^2 + \omega^2)^2}$$

$$= \frac{2\omega s}{(s^2 + \omega^2)^2}, \quad s > 0$$

$$\sin(\omega t)$$

$$\frac{\omega}{s^2 + \omega^2} \text{ for } s > 0$$

Laplace Transform of derivative, $\mathcal{L}\left\{\frac{dy}{dt}\right\}$

$$1) \quad \mathcal{L}\left\{\frac{dy}{dt}\right\} = \int_0^\infty \frac{dy}{dt} e^{-st} dt$$

Assuming $s > 0$

$$= \left[y(t) e^{-st} \right]_0^\infty + s \int_0^\infty y(t) e^{-st} dt$$

$$= [0 - y(0)] + s \mathcal{L}\{y\}$$

$$\mathcal{L}\left\{\frac{d}{dt} y\right\} = s \mathcal{L}\{y\} - y(0)$$

$$\frac{dy}{dt}$$

$$sY(s) - y(0)$$

$$\frac{d^2 y}{dt^2}$$

$$s^2 Y(s) - sy(0) - \frac{dy}{dt}(0)$$

D

I

$$+ e^{-st}$$

$$\frac{dy}{dt}$$

$$- (-se^{-st})$$

$$y(t)$$

$$2) \quad \mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} = \mathcal{L}\left\{\frac{d}{dt} \left(\frac{dy}{dt}\right)\right\}$$

$$= s \mathcal{L}\left\{\frac{d}{dt} y\right\} - \frac{dy}{dt}(0)$$

$$= s [s \mathcal{L}\{y\} - y(0)] - \frac{dy}{dt}(0)$$

$$\mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} = s^2 \mathcal{L}\{y\} - sy(0) - \frac{dy}{dt}(0)$$

Laplace Transform 1st order ODE

Example 1:

Solve $\frac{dy}{dt} - 2y = 1$ with $y(0) = 0$

$$\frac{dy}{dt} - 2y = 1$$

$$\mathcal{L}\left\{\frac{dy}{dt} - 2y\right\} = \mathcal{L}\{1\}$$

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} - 2\mathcal{L}\{y\} = \mathcal{L}\{1\}$$

$$[s\mathcal{L}\{y\} - y(0)] - 2\mathcal{L}\{y\} = \frac{1}{s}$$

$$sY - 2Y = \frac{1}{s}$$

$$Y = \frac{1}{s(s-2)}$$

$$\mathcal{L}\{y\} = \frac{1}{2}\left[\frac{1}{s-2} - \frac{1}{s}\right]$$

$$= \frac{1}{2}\left(\frac{1}{s-2}\right) - \frac{1}{2}\left(\frac{1}{s}\right)$$

$$y = \mathcal{L}^{-1}\left\{\frac{1}{2}\left(\frac{1}{s-2}\right) - \frac{1}{2}\left(\frac{1}{s}\right)\right\}$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$$

$$= \frac{1}{2}[e^{2t}] - \frac{1}{2}[1]$$

$$y = \frac{1}{2}e^{2t} - \frac{1}{2} \quad \#$$

Example 2

Solve $3\frac{dy}{dt} + y = t^2$ with $y(0) = 1$

$$3\mathcal{L}\left\{\frac{dy}{dt}\right\} + \mathcal{L}\{y\} = \mathcal{L}\{t^2\}$$

$$3[sY - y(0)] + Y = \frac{2}{s^3}$$

$$3[sY - 1] + Y = \frac{2}{s^3}$$

$$Y(3s + 1) = \frac{2}{s^3} + 3$$

$$Y = \frac{2}{s^3(3s+1)} + \frac{3}{3s+1}$$

∴ Partial Fraction

$$\mathcal{L}\{y\} = \frac{18}{s} - \frac{6}{s^2} + \frac{2}{s^3} - \frac{51}{3s+1}$$

$$y = \mathcal{L}^{-1}\left\{\frac{18}{s} - \frac{6}{s^2} + \frac{2}{s^3} - \frac{51}{3s+1}\right\}$$

$$= 18\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 6\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} - 17\mathcal{L}^{-1}\left\{\frac{1}{s+\frac{1}{3}}\right\}$$

$$y = 18[1] - 6[t] + [t^2] - 17e^{\frac{1}{3}t}$$

$$y = -17e^{\frac{1}{3}t} + t^2 - 6t + 18 \quad \#$$

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$\cosh(\lambda t)$	$\frac{s}{s^2 - \lambda^2}$ for $s > \lambda $
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$ for $s > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$ for $s > 0$
t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
t^α for $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$ for $s > 0$
$g(t)e^{at}$	$G(s-a)$
$tg(t)$	$-\frac{d}{ds}(G(s))$
$t^n g(t)$	$(-1)^n \frac{d^n}{ds^n}(G(s))$
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$\frac{d^2y}{dt^2}$	$s^2Y(s) - sy(0) - \frac{dy}{dt}(0)$
$\int_0^t g(\tau) d\tau$	$\frac{1}{s}G(s)$
$g(t-a)u(t-a)$	$e^{-as}G(s)$
$\delta(t-a)$	e^{-as} for $a \geq 0$
$(g * h)(t) = \int_0^t g(\tau)h(t-\tau) d\tau$	$G(s)H(s)$

Laplace Transform 2nd order ODE

Example 1

Solve $\frac{d^2 y}{dt^2} + 4y = 0$ with $y(0) = 1$, $\frac{dy}{dt}(0) = -1$

$$\mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} + 4\mathcal{L}\{y\} = 0$$

$$[s^2 Y - sy(0) - \frac{dy}{dt}(0)] + 4Y = 0$$

$$(s^2 + 4)Y - s(1) - (-1) = 0$$

$$Y = \frac{s-1}{s^2+4}$$

$$y = \mathcal{L}^{-1}\left\{\frac{s-1}{s^2+4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} - \frac{1}{2}\mathcal{L}\left\{\frac{2}{s^2+2^2}\right\}$$

$$y = \cos(2t) - \frac{1}{2}\sin(2t)$$

Example 2

Solve $\frac{d^2 y}{dt^2} - 3\frac{dy}{dt} + 2y = e^{-t}$ with $y(0) = -1$, $\frac{dy}{dt}(0) = 0$

$$\mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} - 3\mathcal{L}\left\{\frac{dy}{dt}\right\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\}$$

$$[s^2 Y - sy(0) - \frac{dy}{dt}(0)] - 3[sY - y(0)] + 2Y = \frac{1}{s+1}$$

$$(s^2 - 3s + 2)Y + s - 3 = \frac{1}{s+1}$$

$$Y = \frac{1}{(s+1)(s-1)(s-2)} + \frac{3-s}{(s-1)(s-2)}$$

∴ Partial Fraction

$$\mathcal{L}\{y\} = \frac{1}{6} \frac{1}{s+1} + \frac{4}{3} \frac{1}{s-2} - \frac{5}{2} \frac{1}{s-1}$$

$$y = \frac{1}{6} \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + \frac{4}{3} \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \frac{5}{2} \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$= \frac{1}{6} [e^{-t}] + \frac{4}{3} [e^{2t}] - \frac{5}{2} [e^t]$$

$$y = \frac{1}{6} e^{-t} + \frac{4}{3} e^{2t} - \frac{5}{2} e^t$$

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$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$ for $s > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$ for $s > 0$
t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
t^α for $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$ for $s > 0$
$g(t) e^{at}$	$G(s-a)$
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$\frac{dy}{dt}$	$sY(s) - y(0)$
$\frac{d^2 y}{dt^2}$	$s^2 Y(s) - sy(0) - \frac{dy}{dt}(0)$
$\int_0^t (g(\tau)) d\tau$	$\frac{1}{s} G(s)$
$g(t-a) u(t-a)$	$e^{-as} G(s)$
$\delta(t-a)$	e^{-as} for $a \geq 0$
$(g * h)(t) = \int_0^t (g(\tau) h(t-\tau)) d\tau$	$G(s) H(s)$